UTKRISHTA 2026 | SSG × RESEARCH CLUB, IIIT SRI CITY

LFM Radar Pulse Compression & +10 dB SNR Boost for Long-Range Space Target Tracking

Solving the Radar Energy-Resolution Dilemma at 200 km Range with Linear Frequency Modulation

TRACK 1: Radar Signal Processing — Target Detection, Pulse Compression & SNR Enhancement
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Team: Spaceship
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Problem Overview & Space-Tracking System Setup

The Long-Range Radar Dilemma

• Mission Goal: Detect distant, small RCS space objects (R = 200 km, RCS = 1.0 m²) for launch tracking at SDSC SHAR.

• Energy vs. Resolution Conflict: Higher range demands higher pulse energy $E_t = P_t \cdot \tau$. For unmodulated rectangular pulses, range resolution is tied to pulse duration: $\Delta R = c \cdot \tau / 2$.

• Severe Resolution Degradation: Widening an unmodulated pulse to 100 μs destroys resolution to 15.0 km (unusable for orbital tracking).

• The LFM Solution: Modulating carrier frequency across B = 4 MHz achieves $\Delta R = c/(2B)$ = 37.5 m, completely invariant to pulse duration!

Assumed S-band Radar Parameters

• Carrier Frequency (f_c): 3.0 GHz (S-Band, λ = 0.1 m)

• Pulse Repetition Time (PRT): 2.0 ms (PRF = 500 Hz, R_{unamb} = 300 km)

• Chirp Sweep Bandwidth (B): 4.0 MHz (Nominal ΔR = 37.5 m)

• Pulse Widths Compared: 10 μs (TB = 40) vs 100 μs (TB = 400)

• Transmit Peak Power (P_t): 250 kW (53.98 dBW)

• Antenna Gain (G_{tx}, G_{rx}): 40.0 dBi each (Beamwidth ~ 1.8°)

• Target Range & RCS: R = 200.0 km, RCS = 1.0 m^2 (0 dBsm)

• Received Power (P_r): 3.95 × 10^{-14} W (-134.04 dBW, sub-noise)

• Noise Figure & Losses: F = 2.5 dB, System Loss L = 3.0 dB

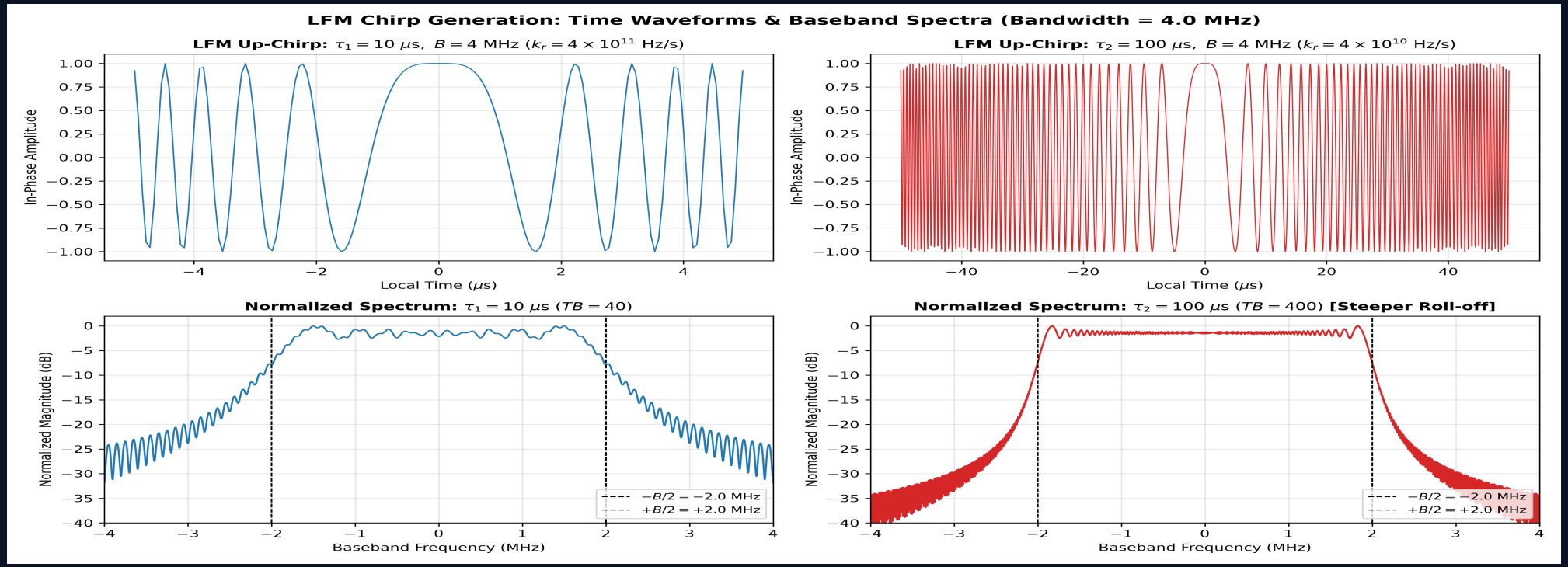
Simulation Methodology & Mathematical Pipeline

Step 1: Signal Synthesis
PRT = 2 ms, B = 4 MHz, fs = 16 MHz. Dual LFM up-chirps: s(t) = rect(t/τ) exp(j π kr t^2). Slope kr1 = 4e11 Hz/s vs kr2 = 4e10 Hz/s.

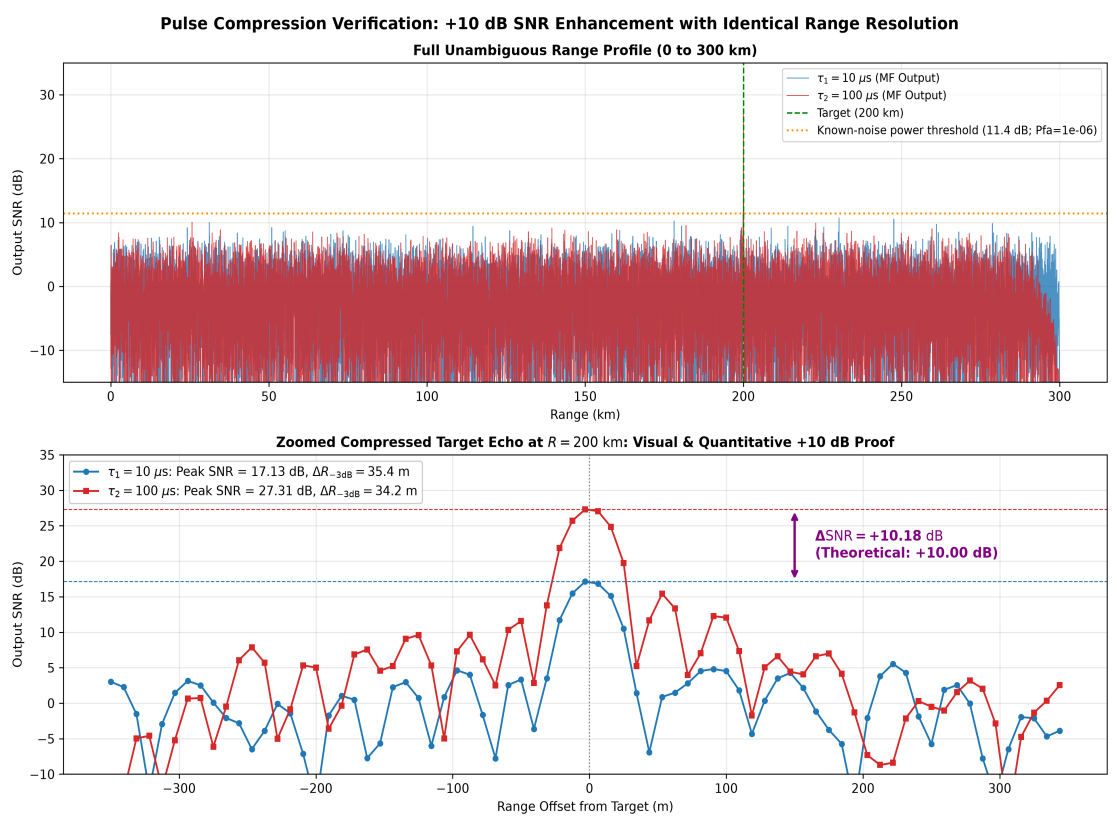
Step 2: Channel & AWGN
Radar Range Equation at R = 200 km (delay 1.333 ms). Circular complex AWGN with discrete-time variance Pn = N0 · fs.

Step 3: Matched Filter (MF)
Zero-padded FFT linear cross-correlation: y(t) = ifft(fft(rx) · conj(fft(ref))). Trims circular convolution artifacts.

Step 4: Detection & CFAR
1-D Cell-Averaging CFAR (Nref=48, Nguard=16, Pfa=1e-6). Link margin verification against Swerling 0 Marcum Q model.



Quantitative Verification & Experimental Plots



Parameter / Metric	Pulse 1 (10 µs)	Pulse 2 (100 µs)	Gain / Ratio
Time-Bandwidth (TB)	40 (16.0 dB)	400 (26.0 dB)	**+10.0 dB**
Radiated Energy	2.50 J	25.00 J	**+10.0 dB**
Theoretical MF SNR	17.44 dB	27.44 dB	**+10.00 dB**
Simulated Peak SNR	17.13 dB	27.31 dB	**+10.18 dB (Valid!)**
Nominal Resolution	37.50 m	37.50 m	**1.00x (Identical)**
Measured -3dB Width	35.36 m	34.17 m	**Similar widths**
Required SNR (Pd=0.9)	13.18 dB	13.18 dB	Marcum Q baseline
Theoretical Link Margin	+4.25 dB (Marginal)	+14.25 dB (Robust)	**+10.00 dB Boost**
CA-CFAR Target Margin	+4.14 dB	+10.08 dB	**+5.93 dB Boost**

Insights, Bonus Experiments & Conclusions

Operational Insights & Bonus Experiments

• Core Proof Verified: Matched-filter SNR increases by +10.18 dB (theory: +10.00 dB) while range resolution remains invariant at ~35 m for both 10 µs and 100 µs chirps.

• Validation: 200 production trials: mean 9.99 dB; 95% mean CI [9.88, 10.10] dB. Assumed link parameters; correlated MF cells limit nominal CFAR Pfa calibration.

• Bonus Exp 1 (Multi-Target at 200 km): The 50 m pair yields one local peak in this coherent unequal-target realization. Nominal c/(2B) resolution alone does not guarantee separation; phase, contrast and noise matter.

• Bonus Exp 2 (Mach 5 Doppler Coupling): A 1500 m/s rocket induces -112.5 m range bias. We implemented Chirp Diversity (Up/Down chirp pairs), canceling the bias to -3 m and extracting exact velocity (1500 m/s)!

